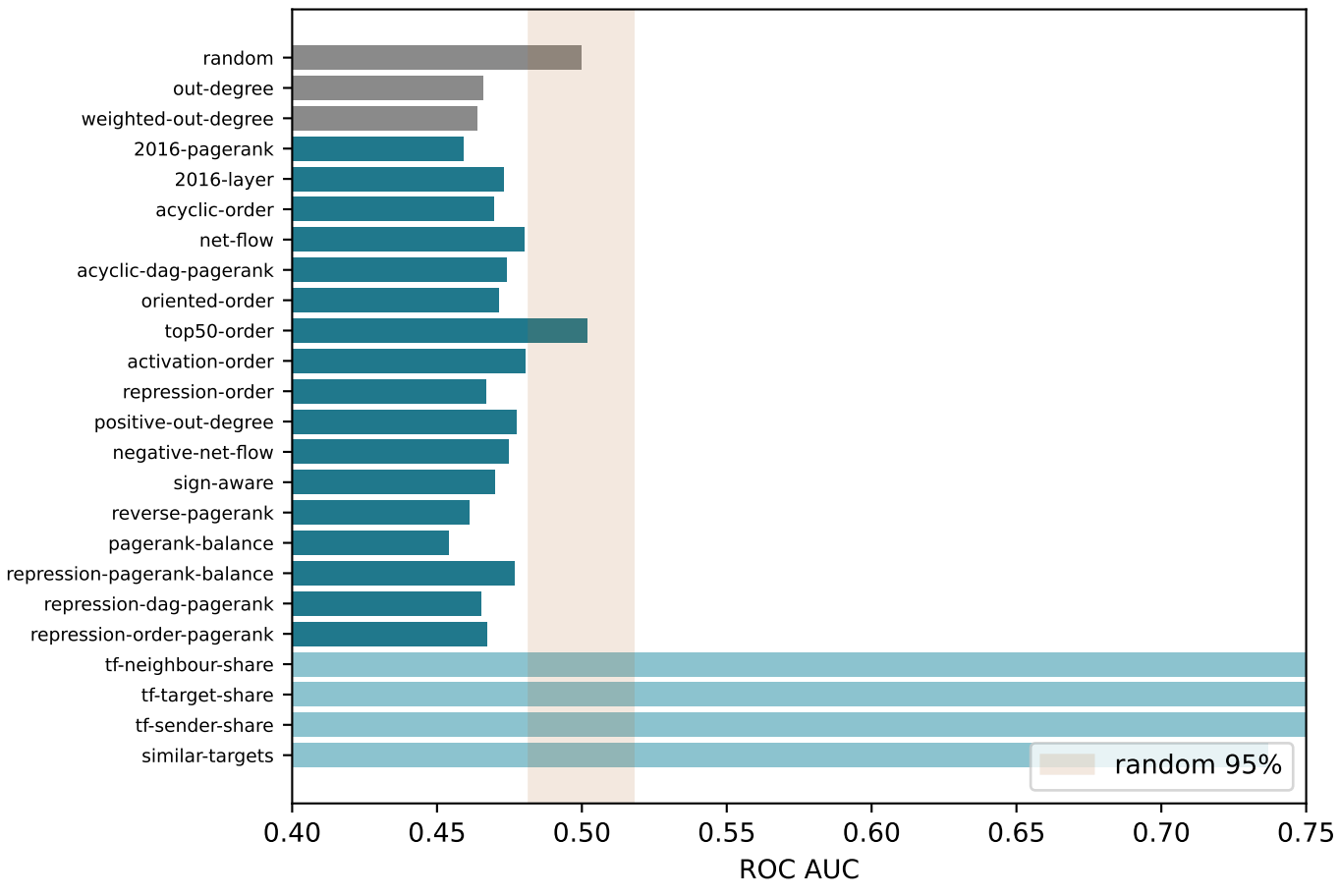
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,916 genes ranked, 1,147 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.7 (4–16)	0.500 (0.482–0.518)				
out-degree	5	0.466	0.962	1.000	+0.000 (+0.000, +0.000)	
weighted-out-degree	6	0.464	0.922	1.000	-0.002 (-0.003, -0.001)	< 0.001
2016-pagerank	8	0.459	0.767	1.000	-0.007 (-0.024, +0.012)	0.512
2016-layer	8	0.473	0.767	0.999	+0.007 (-0.010, +0.024)	0.486
acyclic-order	5	0.469	0.962	1.000	+0.004 (-0.003, +0.010)	0.268
net-flow	5	0.480	0.962	0.984	+0.014 (+0.005, +0.024)	< 0.001
acyclic-dag-pagerank	6	0.474	0.922	0.999	+0.008 (+0.003, +0.013)	0.002
oriented-order	5	0.471	0.962	1.000	+0.005 (-0.002, +0.014)	0.152
top50-order	14	0.502	0.099	0.429	+0.036 (+0.017, +0.055)	0.004
activation-order	9	0.480	0.657	0.983	+0.014 (+0.003, +0.025)	0.014
repression-order	4	0.467	0.983	1.000	+0.001 (-0.019, +0.020)	0.900
positive-out-degree †	7	0.477	0.856	0.996	+0.012 (+0.004, +0.018)	0.002
negative-net-flow †	5	0.475	0.962	0.998	+0.009 (-0.008, +0.024)	0.288
sign-aware †	11	0.470	0.377	1.000	+0.004 (-0.004, +0.012)	0.342
reverse-pagerank †	6	0.461	0.922	1.000	-0.005 (-0.008, -0.001)	0.006
pagerank-balance †	5	0.454	0.962	1.000	-0.012 (-0.023, -0.000)	0.048
repression-pagerank-balance †	7	0.477	0.856	0.996	+0.011 (-0.003, +0.026)	0.122
repression-dag-pagerank †	7	0.465	0.856	1.000	-0.001 (-0.021, +0.019)	0.962
repression-order-pagerank †	6	0.467	0.922	1.000	+0.001 (-0.019, +0.020)	0.900
tf-neighbour-share ‡	93	0.769	< 0.001	< 0.001	+0.303 (+0.280, +0.327)	< 0.001
tf-target-share ‡	73	0.764	< 0.001	< 0.001	+0.298 (+0.275, +0.323)	< 0.001
tf-sender-share ‡	90	0.758	< 0.001	< 0.001	+0.292 (+0.269, +0.315)	< 0.001
similar-targets ‡	78	0.737	< 0.001	< 0.001	+0.271 (+0.247, +0.294)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

collectri

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,864	0.047 (2.83×)	3.43×	1.30×	0.78 (0.77)
resolve_reciprocals → drop_sinks → prune_triangles	31,844	0.006 (1.28×)	3.38×	1.13×	0.29 (0.29)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	17,074	0.006 (0.77×)	4.16×	0.73×	0.21 (0.21)
absolute	31,864	0.047 (2.83×)			
absolute → minimum_feedback_order	31,698	0.030 (2.38×)			
orient_reciprocals ratio=2.0 → absolute	31,737	0.020 (2.02×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	31,645	0.016 (1.78×)			
top_outgoing k=50 → absolute	31,864	0.019 (5.02×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,613	0.012 (4.51×)			
keep_sign sign=1 → absolute	31,855	0.041 (3.42×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,405	0.028 (3.14×)			
keep_sign sign=-1 → absolute	23,620	0.008 (1.20×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	20,437	0.007 (1.04×)			
keep_sign sign=-1	23,620	0.008 (1.20×)		1.20×	0.18 (0.18)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,491,299 edges (720,311 negative), largest strong component 11,913.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,491,299 edges (720,311 negative), largest strong component 11,913.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 704,512 edges (662,384 negative), largest strong component 11,702.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 704,512 edges (662,384 negative), largest strong component 11,702; DAG keeps 566,729 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,491,299 edges (0 negative), largest strong component 11,913; DAG keeps 1,828,517 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,491,299 edges (0 negative), largest strong component 11,913.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,491,299 edges (0 negative), largest strong component 11,913; DAG keeps 1,828,517 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,487,402 edges (0 negative), largest strong component 11,809; DAG keeps 1,338,177 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 527,333 edges (0 negative), largest strong component 11,911; DAG keeps 357,547 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,770,988 edges (0 negative), largest strong component 11,854; DAG keeps 1,240,519 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 720,311 edges (0 negative), largest strong component 7,600; DAG keeps 667,996 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,491,299 edges (720,311 negative), largest strong component 11,913.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,491,299 edges (720,311 negative), largest strong component 11,913.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,491,299 edges (720,311 negative), largest strong component 11,913.

reverse-pagerank: Highest reverse PageRank (raw graph)